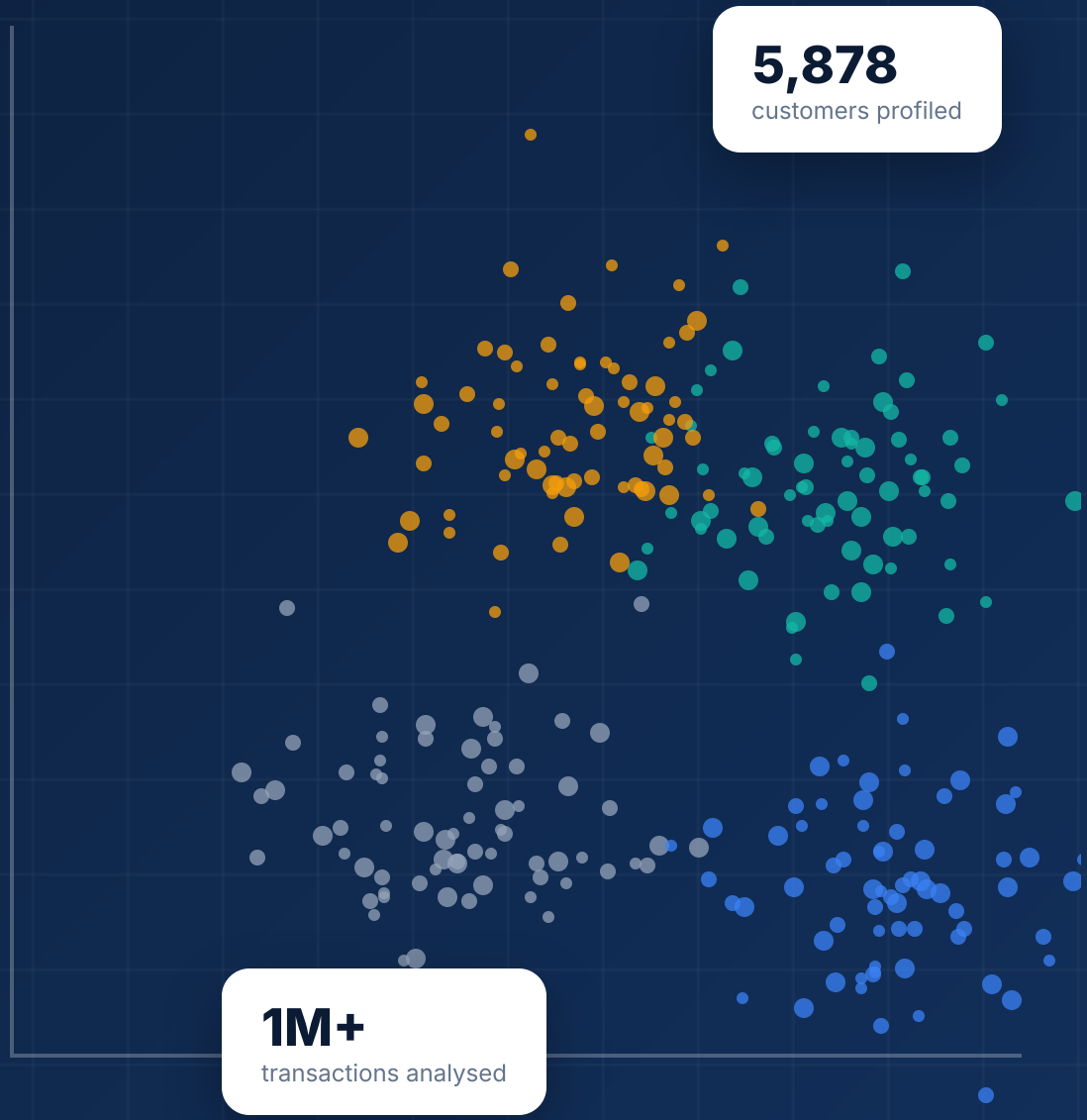
CASE STUDY

E-commerce Customer Segmentation & RFM Analysis

From 1M+ transactions to actionable customer insights

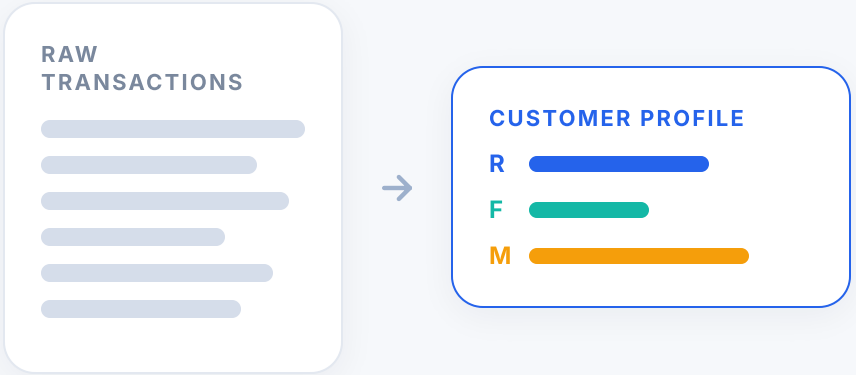
Python • SQL • Machine Learning • Power BI



02 · BUSINESS PROBLEM

From Transactions to Customer Insights

E-commerce businesses generate large volumes of transaction data, but raw transactions do not directly explain customer behavior.



01

Who are the most valuable customers?

02

Which customers are at risk?

03

Which customers show strong loyalty?

04

How can customer groups be identified for targeted decisions?

From Raw Data to Business Intelligence

END-TO-END WORKFLOW

DATA PREPARATION

1 1M+ Transactions
Original dataset

2 Python Data Cleaning
Cancelled invoices · Returns · Invalid prices · Missing IDs · Duplicates · Revenue created · Final validation

3 779K+ Valid Transactions
36,969 invoices

4 5,878 Customers
Customer-level view

CUSTOMER ANALYTICS

5 RFM Analysis
Recency · Frequency · Monetary

6 Customer Segmentation
7 rule-based RFM segments

7 K-Means Clustering
4 data-driven clusters

BUSINESS DELIVERY

8 SQL Business Analysis
MySQL · KPIs · validation

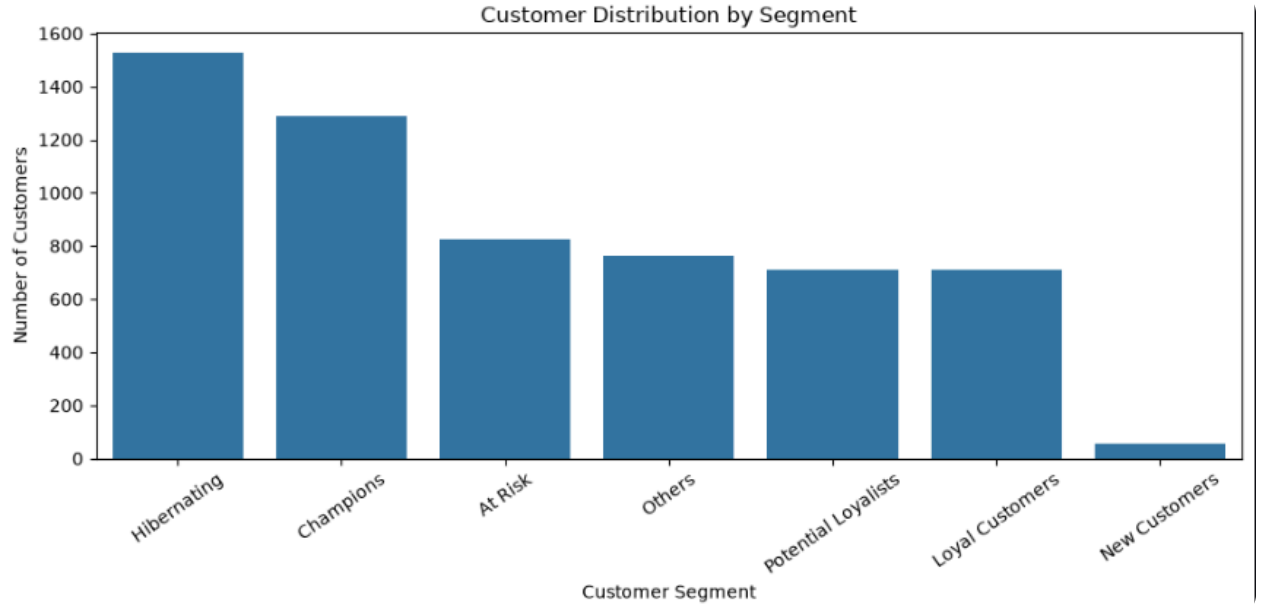
9 Power BI Dashboard
4-page interactive dashboard

Understanding Customer Value with RFM

R **Recency**
How recently a customer purchased

F **Frequency**
How often a customer purchased

M **Monetary**
How much a customer spent



Customer distribution by RFM segment · from my Jupyter Notebook

KEY CUSTOMER SEGMENTS

Champions
Recent, frequent, high spend

Loyal Customers
Frequent buyers, strong spend

Potential Loyalists
Recent buyers, building frequency

New Customers
Very recent, single order

At Risk
Previously valuable, now inactive

Hibernating
Inactive, low purchase activity

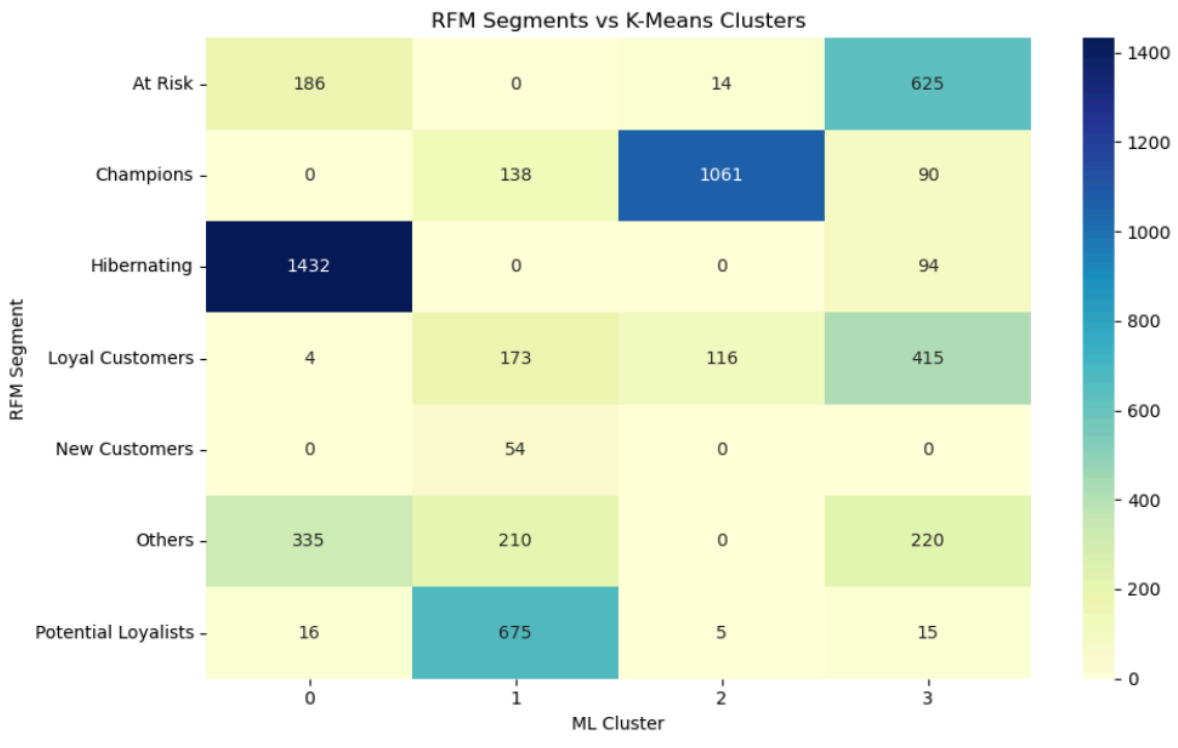
Adding a Machine Learning Perspective

RFM Features → Log Transformation → StandardScaler → K-Means → Silhouette Score → **4 Customer Clusters**

CLUSTER INTERPRETATION

Cluster 0	**Hibernating**
Cluster 1	**Recent**
Cluster 2	**High Value**
Cluster 3	**At Risk**

> "K-Means provided a data-driven perspective that could be compared with the business-rule-based RFM segmentation."



RFM segments vs K-Means clusters · from my Jupyter Notebook

Turning Customer Data into Business Questions

MySQL supported the analysis from database setup through business analysis.

- **Overall business KPIs**
- **Monthly revenue analysis**
- **Customer analysis**
- **RFM-related analysis**
- **Business analysis queries**
- **Data validation**

Database creation · Table creation · Data loading

MySQL Workbench

```sql
-- 4. Top 10 Customers by Purchase Frequency

SELECT
    `Customer ID`,
    Recency,
    Frequency,
    ROUND(Monetary, 2) AS Monetary,
    RFM_Score,
    Segment
FROM customer_rfm
ORDER BY Frequency DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Customer ID	Recency	Frequency	Monetary	RFM_Score	Segment
17841	2	211	68545.25	555	Champions
15311	1	208	114966.42	555	Champions
13089	3	203	113416.91	555	Champions
14606	1	192	29951.37	555	Champions
14156	10	156	313437.62	555	Champions
17850	372	155	51208.87	255	At Risk
14646	2	151	528602.52	555	Champions

Top 10 customers by purchase frequency, queried from the customer_rfm table

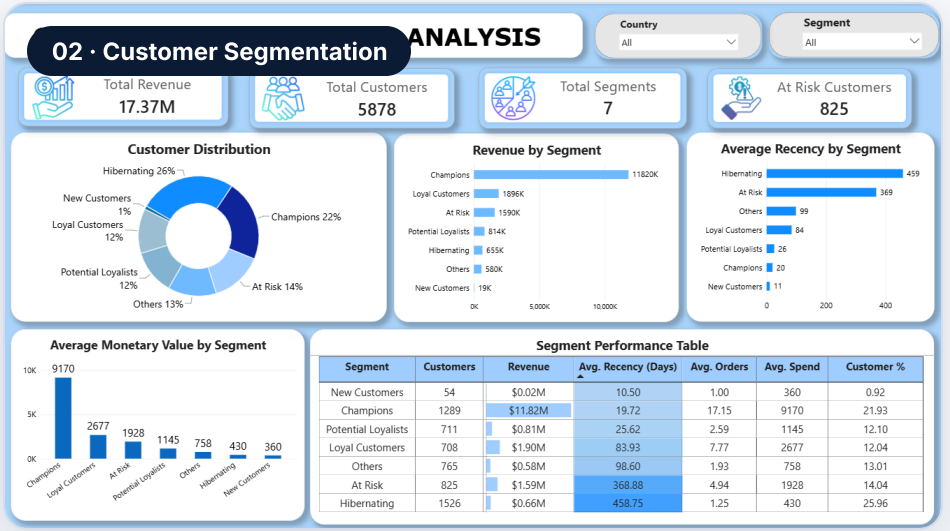
07 · POWER BI

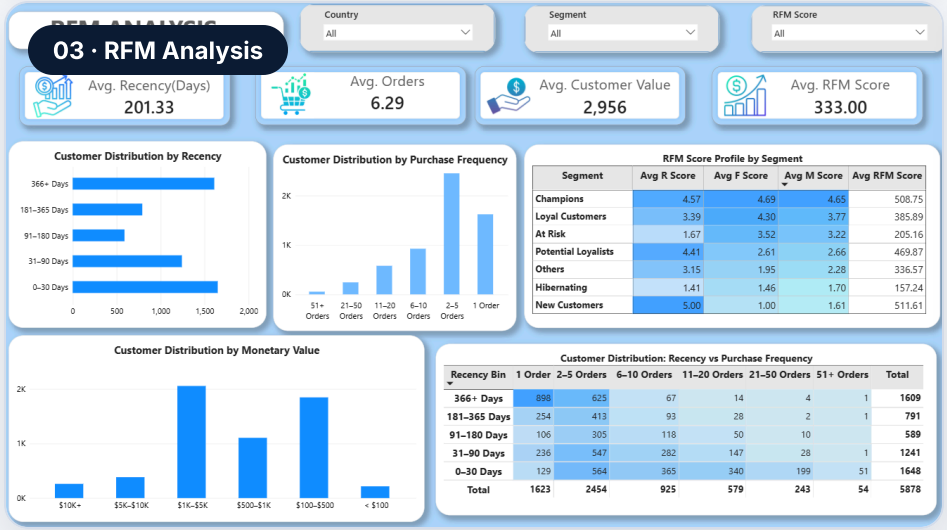
Interactive Power BI Dashboard

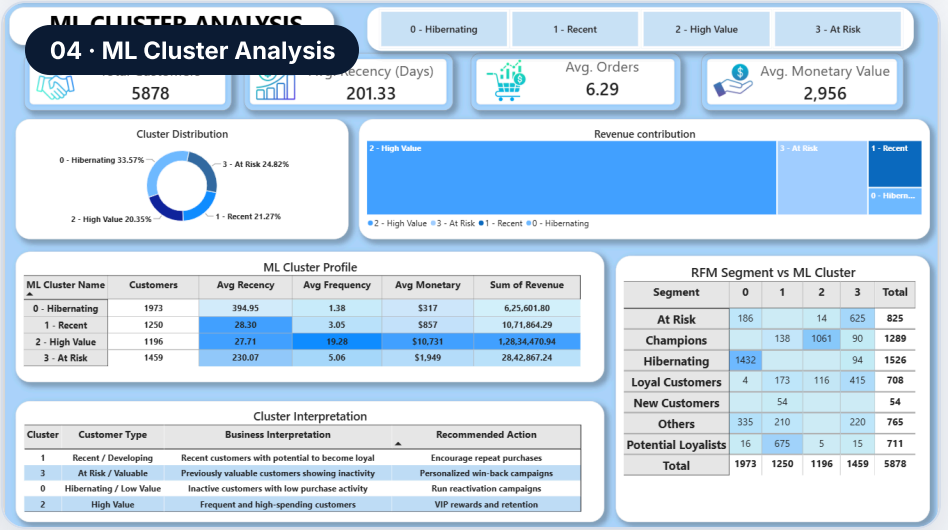
4 completed report pages

1. Sales Overview
2. Customer Segmentation
3. RFM Analysis
4. ML Cluster Analysis









08 · KEY INSIGHTS

What the Data Showed

	INSIGHT	BUSINESS IMPLICATION
Nov peak month	01 · SEASONAL SALES PEAK **November recorded the highest monthly sales in both 2010 and 2011.**	→ Prepare inventory, promotions and marketing campaigns ahead of this recurring high-demand period.
~68% of total revenue	02 · REVENUE CONCENTRATION **Champions are 22% of customers (1,289) but generate $11.82M, approximately 68% of total revenue.**	→ Protect and retain this high-value group to maintain revenue.
$1.59M from At Risk	03 · WIN-BACK OPPORTUNITY **825 At Risk customers (~14% of the base) generated $1.59M, with an average recency of 368.88 days.**	→ Their historical value makes them a target for personalized re-engagement and win-back campaigns.
$1.90M from Loyal	04 · LOYALTY & RETENTION **708 Loyal Customers generate approximately $1.90M in revenue.**	→ Loyalty rewards and personalized offers can help maintain and strengthen this group.
4 ML clusters	05 · ML COMPLEMENTARY VIEW **K-Means identified 4 interpretable behavioral clusters. The High Value cluster largely aligns with RFM Champions.**	→ Combine RFM with unsupervised clustering for multiple perspectives on customer behavior.

09 · TECH STACK

Tools Behind the Analysis

ANALYSIS

- Python
- Pandas
- NumPy

MACHINE LEARNING

- Scikit-learn

VISUALIZATION

- Matplotlib
- Seaborn
- Power BI

DATA & SQL

- MySQL
- SQL
- Excel

WORKFLOW

- Jupyter Notebook
- Git
- GitHub

WHAT THIS PROJECT DEMONSTRATES

End-to-end workflow
Raw data to dashboard

Data cleaning
Python · Pandas

Segmentation
RFM + K-Means

SQL & BI reporting
MySQL · Power BI